Can Semantic Similarity Capture Headedness Distributions? Evidence from Chinese Disyllabic Compounds

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Mandarin Chinese compounds are central to Chinese word formation, yet their heads are often difficult to identify from constituent order or lexical category alone. This study asks whether headedness patterns in Chinese disyllabic compounds are stable across lexical samples and whether they can be approximated through compound–constituent semantic similarity. Experiment 1 analyzes 5,024 high-frequency compounds and broadly replicates the main class–headedness mapping of Ceccagno and Basciano (2007, 2008). Experiment 2 uses word embeddings to test whether these patterns are reflected in distributional semantic space. We define two complementary scores: the semantic asymmetry score (Δ), which measures left–right imbalance in compound–constituent similarity, and the semantic proximity score (τ), which measures anchoring in the closest constituent. We find that Δ captures headedness direction most clearly for right-headed compounds and more weakly for left-headed and two-headed compounds, while τ captures the reduced constituent anchoring of exocentric compounds. These findings suggest that Chinese compound headedness is structurally conditioned and reflected as gradient patterns in semantic space.

1. Introduction

Compounding occupies a central position in Mandarin Chinese word formation. A large proportion of Chinese lexical items are compounds, and disyllabic compounds constitute a particularly important part of the modern lexicon (Ceccagno & Basciano 2007; Packard 2000; Zhou et al. 1999). This makes Chinese a useful testing ground for theories of compound structure, especially for the notion of morphological headedness. In general morphology, the head of a complex word is usually understood as the constituent that determines the syntactic category and provides the core semantic type of the whole word (Lieber 2004; Williams 1981). In a canonical endocentric compound such as *dog bed*, the compound denotes a subtype of the right-hand constituent *bed*. The question for Chinese is whether this notion can be applied systematically to compounds whose constituents are often category-flexible, semantically relational, and highly lexicalized.

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Previous studies have given different answers to this question. Some analyses describe Chinese compounds as largely right-headed (Starosta et al. 1998), while Packard (2000) proposes a more differentiated tendency: nominal compounds are generally right-headed, whereas verbal compounds are often left-headed. A stronger position is found in Huang (1998), who argues that Chinese compounding is fundamentally headless, since the syntactic category or semantic type of many compounds cannot be derived from a single constituent. These disagreements reflect a deeper empirical difficulty. In Chinese compounds, categorial headedness, semantic headedness, and syntactic behavior do not always coincide. The same surface construction may yield different compound categories, and a constituent may determine formal category while the meaning of the whole compound is metaphorical or external to both constituents. Headedness in Chinese therefore has to be treated as an empirical property of individual compounds, rather than as a simple consequence of constituent order.

A major step in this direction is the framework developed by Ceccagno and Basciano (2007, 2008), building on the cross-linguistic compound typology of Bisetto and Scalise (2005) and later work on compound classification (Scalise & Bisetto 2009). In this framework, compounds are classified according to the grammatical relation between their constituents. Subordinate compounds involve an argument–head relation, attributive compounds involve a modifier–head relation, and coordinate compounds combine two constituents of comparable semantic or categorial status. Ceccagno and Basciano further assign Chinese disyllabic compounds to four headedness patterns: left-headed, right-headed, two-headed, and exocentric. This approach shifts the problem from asking whether Chinese is uniformly left-headed, right-headed, or headless to asking how different headedness patterns are distributed across structural classes. Their neologism-based study shows clear regularities: attributive compounds are predominantly right-headed, subordinate compounds tend to be left-headed, coordinate compounds are characteristically two-headed, and exocentric compounds are comparatively less frequent (Ceccagno & Basciano 2007, 2008).

Two issues remain open. The first concerns empirical stability. Ceccagno and Basciano's original study is based on 672 neologisms, a dataset well suited for studying productive word formation but potentially shaped by register, historical period, media usage, and patterns of lexical innovation. A broader high-frequency lexicon may contain more established and entrenched compounds, and it is unclear in advance whether the same headedness distributions will hold. The second issue concerns measurement. Existing work on Chinese compound headedness relies mainly on categorical linguistic annotation. Such annotation is indispensable, but if headedness reflects the relative semantic contribution of constituents, it may also leave gradient traces in semantic representation. This raises the possibility of testing whether categorically annotated headedness patterns correspond to measurable patterns of constituent–compound semantic similarity.

This possibility is supported by work on semantic transparency and distributional semantics. In psycholinguistic research, compound transparency is often treated as a constituent-sensitive property: the meanings of the whole word and of its individual morphemes may remain closely related, partially related, or opaque (Libben et al. 2003; Schäfer 2018). Computational studies have further shown that constituent–compound semantic similarity can be modeled using vector-based representations, including latent semantic analysis and word embeddings (Bell & Schäfer 2013, 2016; Wang et al. 2014). These methods are especially relevant for Chinese, where recent work emphasizes that compounding is strongly shaped by semantic relations among constituents (Song et al. 2022). More generally, distributional semantic models represent word meaning in terms of contextual co-occurrence and allow semantic relatedness to be quantified continuously, often by cosine similarity between vectors (Baroni & Lenci 2010; Landauer & Dumais 1997; Turney & Pantel 2010). Such measures provide a way to test whether headedness has a detectable usage-based correlate in semantic space.

The present study addresses these issues in two experiments. Experiment 1 tests whether the headedness distributions reported by Ceccagno and Basciano (2007, 2008) can be replicated in a larger lexicon-based dataset of 5,024 high-frequency Chinese disyllabic compounds drawn from the BCC corpus (Xun et al. 2016). This experiment evaluates whether the structural-class and headedness system is stable beyond neologism data. Experiment 2 then asks whether semantic similarity can approximate these manually assigned headedness patterns. Using Tencent Chinese word embeddings (Song et al. 2018), we compute the cosine similarity between each compound and its left and right constituents. From these values, we define two complementary measures: the semantic asymmetry score (Δ), which captures the direction and magnitude of the imbalance in constituent–compound similarity, and the semantic proximity score (τ), which captures the degree to which the compound remains semantically anchored in at least one constituent. If headedness leaves a distributional-semantic trace, left-headed compounds should tend toward positive Δ values, right-headed compounds toward negative Δ values, and two-headed compounds toward values near zero. If exocentric compounds are less strongly anchored in either constituent, they should show lower semantic proximity scores (τ) than endocentric compounds on average. The goal is to complement categorical morphological analysis with a quantitative semantic test. By first replicating headedness distributions and then testing whether they are reflected in semantic similarity, this study evaluates whether Chinese compound headedness is both structurally stable and reflected as a gradient pattern in distributional semantic space.

2. Experiment 1: Replication Analysis

Experiment 1 tests whether the headedness distributions reported by Ceccagno and Basciano (2007, 2008) extend beyond their original neologism corpus. This step is necessary for the logic of the present study. If the association between structural class and headedness were limited to a particular neologism dataset, it would provide a weak target

for the semantic similarity analyses in Experiment 2. A successful replication in a broader high-frequency lexicon would suggest that the classification captures stable properties of Chinese compound structure.

2.1. DATA AND ANNOTATION. The dataset was constructed from the frequency list of commonly used Chinese words in the BCC corpus (Xun et al. 2016). To focus on established lexical items and reduce noise from rare or specialized forms, we selected high-frequency disyllabic compounds occurring more than 1,000 times in the corpus. Items with unclear morphosyntactic structure were excluded. Since Experiment 2 uses distributional semantic measures, the dataset was also aligned with the embedding vocabulary used in that experiment: compounds absent from the embedding inventory, or compounds whose constituent characters were absent from the embedding inventory, were removed. After filtering, the final dataset contained 5,024 Chinese disyllabic compounds.

Each compound was manually annotated following the framework of Ceccagno and Basciano (2007, 2008), which builds on the relation-based compound typology of Bisetto and Scalise (2005). The annotation included five fields: the compound itself, its structural class, its construction type, its lexical category, and its headedness pattern. Structural class was coded as subordinate (SUB), attributive (ATT), or coordinate (CRD). Headedness was coded as left-headed, right-headed, two-headed, or exocentric. The coding of headedness was based primarily on semantic headedness, with syntactic and categorial information considered when relevant.[1]

Compound	Class	Construction	Category	Head	Gloss
攀高	SUB	V + A	V	left-headed	climb + high = climb upward
开会	SUB	V + N	V	left-headed	hold + meeting = hold a meeting
品读	ATT	V + V	V	right-headed	savor + read = read with appreciation
高考	ATT	A + N	N	right-headed	high + exam = college entrance exam
打压	CRD	V + V	V	two-headed	strike + suppress = crack down on
酸甜	CRD	A + A	A	two-headed	sour + sweet = sweet-and-sour
网虫	SUB	N + N	N	exocentric	net + insect = web enthusiast
花心	ATT	A + N	A	exocentric	flashy + heart = unfaithful

Table 1. Annotated examples and coding scheme.

[1] In this study, headedness is a morphological annotation label, with semantic headedness as the primary criterion. For endocentric compounds, the head is the constituent that provides the semantic type of the whole compound, with categorial or syntactic evidence used as support. Left- and right-headed labels indicate the position of this head; two-headed labels indicate comparable semantic centers in coordinate compounds; exocentric labels indicate that no internal constituent provides the semantic type of the whole word, or that formal category and semantic type do not converge on the same constituent.

A note on terminology is needed. Ceccagno and Basciano (2007) discuss some of their distributions in terms of productivity. In the present study, we use the term frequency distribution for the descriptive comparisons in Experiment 1. In corpus-based morphology, productivity in the strict quantitative sense is usually linked to vocabulary growth and the likelihood of encountering new types, often estimated through low-frequency or hapax-based measures (Baayen 2009). Since the present dataset is deliberately restricted to high-frequency lexical items, the analysis is designed to test distributional stability rather than productivity in that technical sense.

2.2. REPLICATION STRATEGY. The two datasets differ in size and lexical profile: their study analyzes 672 neologisms, whereas the present study analyzes 5,024 high-frequency compounds from a broader lexicon. The relevant question is whether the same structural generalizations hold across these different lexical samples.

Four diagnostics were used for comparison. First, we examined the overall distribution of the three structural classes. Second, we compared the distribution of lexical categories across the dataset. Third, we examined the relation between structural class and headedness pattern, which is the central test for Ceccagno and Basciano's classification. Fourth, we compared the most frequent construction types within each class. Together, these diagnostics test whether the Ceccagno and Basciano system is stable at the levels of class, category, headedness, and construction.

2.3. RESULTS. The overall distribution of structural classes broadly matches the pattern reported by Ceccagno and Basciano (2007). Attributive compounds are the largest class in both datasets. In the present dataset, ATT compounds account for 2,137 items (42.54%), followed by CRD compounds with 1,730 items (34.43%) and SUB compounds with 1,157 items (23.03%). The main difference concerns the relative ordering of SUB and CRD: in the neologism corpus, SUB compounds outnumber CRD compounds, while in the present high-frequency lexicon CRD compounds are more frequent than SUB compounds. This difference plausibly reflects the different lexical profiles of the two datasets. Neologisms are especially useful for observing recent word-formation patterns, whereas a high-frequency lexicon contains many entrenched compounds that may have accumulated across different periods of lexical development.

The distribution of lexical categories is also consistent with the earlier findings. In Ceccagno and Basciano's study, nouns are the most frequent category, followed by verbs and adjectives. The same general ranking is found here: nouns are most frequent (N = 2,363, 47.03%), followed by verbs (V = 1,889, 37.60%), adjectives (A = 547, 10.89%), and adverbs (ADV = 225, 4.48%). The inclusion of adverbs adds a small category absent from some of the original tabulations, but it does not alter the main pattern: nouns and verbs dominate the dataset.

The most important replication concerns the mapping between structural class and headedness. As shown in Figure 1, SUB compounds are predominantly left-headed, ATT compounds are overwhelmingly right-headed, and CRD compounds are overwhelmingly two-headed. Within SUB compounds, left-headed items are the largest group (746 items,

64.50%), followed by right-headed items (239 items, 20.70%) and exocentric items (172 items, 14.90%). Within ATT compounds, right-headed items dominate the class (1,887 items, 88.30%), with exocentric compounds forming a much smaller group (215 items, 10.10%) and left-headed compounds appearing only marginally (35 items, 1.64%). Within CRD compounds, two-headed items are dominant (1,600 items, 92.50%), followed by exocentric items (130 items, 7.51%). No left-headed or right-headed CRD compounds are found in the present dataset.

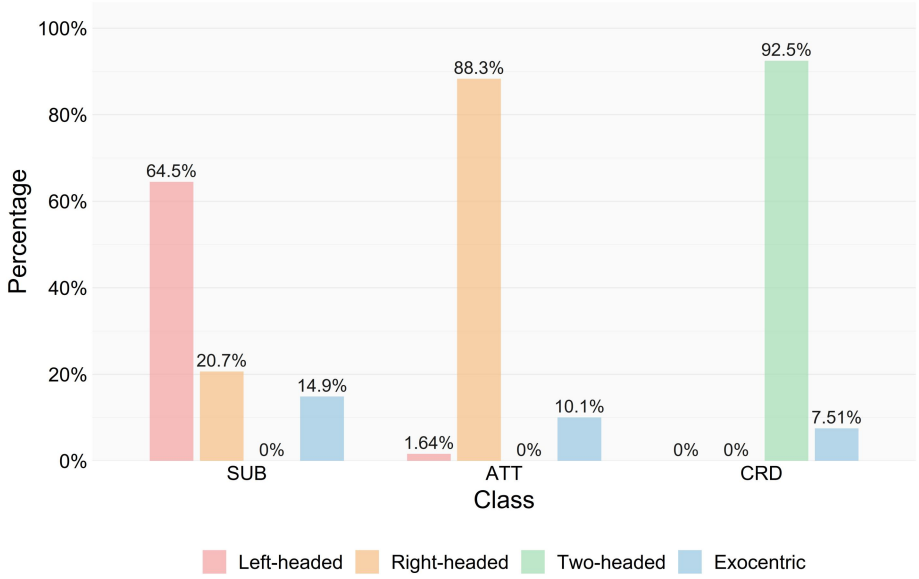


Figure 1. Headedness distribution across structural classes.

This result supports the central claim of the Ceccagno and Basciano framework: headedness in Chinese compounds is systematically associated with structural class. The replication is strongest for ATT and CRD compounds. ATT compounds show a clear modifier–head profile, with the right constituent usually functioning as the semantic head. CRD compounds show the expected two-headed profile, reflecting the comparable contribution of both constituents. SUB compounds are more heterogeneous, but they still show the same dominant tendency found in the original study: left-headed compounds form the largest group.

The category-by-headedness distribution further supports this picture. Left-headed compounds are overwhelmingly verbal: 702 of the 781 left-headed compounds are verbs, accounting for 89.90% of that group. Right-headed compounds are overwhelmingly nominal: 1,654 of the 2,126 right-headed compounds are nouns, accounting for 77.80%. Two-headed compounds are more evenly distributed across verbs, nouns, and adjectives, though verbs remain the largest subgroup. Exocentric compounds show the most mixed categorial profile, with nouns, adjectives, and adverbs all represented. These category

patterns are consistent with the view that headedness in Chinese is shaped jointly by semantic relation, construction type, and lexical category.

The construction-level comparison also largely replicates the earlier study. In SUB compounds, the most frequent construction is $[V+N]_V$, with 353 instances, matching the central role of verb–object and related verbal compounds in this class. In CRD compounds, the dominant constructional hierarchy is $[V+V]_V > [N+N]_N > [A+A]_A$, with 860, 453, and 253 instances respectively. This closely matches the pattern reported for coordinate compounds in the earlier work. The main difference appears in ATT compounds. Ceccagno and Basciano report $[A+N]_N$ as the most frequent attributive pattern, whereas the present dataset has $[N+N]_N$ as the most frequent ATT construction, with 882 instances, followed by $[A+N]_N$ with 452 instances. This difference may reflect the nature of the present dataset: established high-frequency lexical items include many entrenched nominal modifier–head compounds, while neologism corpora may highlight more salient adjective–noun innovation patterns.

Diagnostic	Ceccagno and Basciano	Present dataset	Replicated?
Class distribution	ATT > SUB > CRD	ATT > CRD > SUB	×
Category distribution	N > V > A	N > V > A > ADV	✓
SUB headedness	Left > Right > Exo	Left > Right > Exo	✓
ATT headedness	Right > Exo	Right >> Exo > Left	✓
CRD headedness	Two > Exo > Right	Two > Exo; Right = 0	✓
Top SUB construction	$[V+N]_V$	$[V+N]_V$	✓
Top ATT construction	$[A+N]_N$	$[N+N]_N > [A+N]_N$	×
Top CRD construction	$[V+V]_V > [N+N]_N > [A+A]_A$	$[V+V]_V > [N+N]_N > [A+A]_A$	✓

Table 2. Replication summary against Ceccagno and Basciano (2007, 2008).

Experiment 1 shows that the main distributional generalizations proposed by Ceccagno and Basciano (2007, 2008) are broadly replicated in a larger high-frequency lexicon. The strongest result is the stability of the class-headedness mapping: SUB compounds are mostly left-headed, ATT compounds are mostly right-headed, and CRD compounds are mostly two-headed. Some differences are also observed, especially the higher frequency of CRD compounds relative to SUB compounds and the dominance of $[N+N]_N$ within ATT compounds. These differences are compatible with the contrast between a neologism corpus and a general high-frequency lexicon. Table 2 summarizes the main replicated patterns and the major deviations.

The replication establishes the empirical baseline for Experiment 2. Since the headedness patterns are stable across datasets, they can be used as annotation-based targets for testing whether semantic similarity measures capture the same structural distinctions in distributional semantic space.

3. Experiment 2: Semantic Similarity Analysis

Experiment 2 tests whether the headedness patterns replicated in Experiment 1 are reflected in compound–constituent semantic similarity. The central assumption is that headedness, if it corresponds to the relative semantic contribution of constituents, should leave a measurable trace in distributional semantic space. Specifically, the constituent functioning as the semantic head should tend to be closer to the whole compound than the non-head constituent. Exocentric compounds, by contrast, should show weaker semantic anchoring in either constituent.

3.1. EMBEDDINGS AND SIMILARITY MEASURES. We used Tencent Chinese word embeddings to represent the meanings of compounds and their constituents (Song et al. 2018). Word embeddings represent lexical items as vectors in a high-dimensional semantic space, where words with similar contextual distributions tend to have similar vector representations. In this study, semantic similarity refers specifically to cosine similarity between embedding vectors. Cosine similarity measures the directional similarity between two vectors and ranges from −1 to 1, with higher values indicating greater semantic similarity.

For each disyllabic compound *AB*, where *A* is the left constituent and *B* is the right constituent, we computed the cosine similarity between the compound vector and each constituent vector. This gives two basic measures: sim(*A*), the semantic similarity between *A* and *AB*, and sim(*B*), the semantic similarity between *B* and *AB*. From these two values, we derived two complementary scores.

The first is the semantic asymmetry score (Δ), which captures the direction and magnitude of the imbalance in compound–constituent similarity: $\Delta = \text{sim}(A) - \text{sim}(B)$. A positive Δ value means that the compound is semantically closer to the left constituent than to the right constituent. A negative Δ value means that the compound is closer to the right constituent. A value near zero means that the two constituents are similarly close to the compound. The second is the semantic proximity score (τ), which captures the compound's semantic proximity to its closest constituent: $\tau = \max(\text{sim}(A), \text{sim}(B))$. A high τ value means that the compound remains strongly anchored in at least one of its constituents. A low τ value means that the compound is relatively distant from both constituents.

The compound 高考 "high + exam = college entrance examination" illustrates the logic. 高考 is right-headed because the compound denotes a kind of 考 "exam", with 高 "high" functioning as the modifier. In the embedding space, 高考 is much closer to 考 than to 高: sim(高) = 0.265 and sim(考) = 0.749. This gives $\Delta = -0.484$, indicating a strong semantic asymmetry toward the right constituent, and $\tau = 0.749$, indicating that the compound remains strongly anchored in one constituent.

3.2. PREDICTIONS. The two scores target different aspects of headedness. Δ is designed to capture headedness direction. If a compound is left-headed, the left constituent should be semantically closer to the whole compound, so Δ should tend to be positive. If a compound is right-headed, the right constituent should be closer, so Δ should tend to be

negative. If a compound is two-headed, both constituents should contribute more evenly, so Δ should cluster near zero. τ is designed to capture constituent anchoring. If a compound is endocentric, at least one constituent should remain semantically close to the whole compound. Left-headed, right-headed, and two-headed compounds are therefore expected to show relatively high τ values. If a compound is exocentric, its meaning is expected to be less strongly anchored in either constituent, so exocentric compounds should show lower τ values on average.

For statistical reporting, we report p values and Cohen's d as compact evidence for reliability and effect size. For Δ, we used one-sample t-tests against zero to assess the predicted directional displacement, while interpreting the two-headed near-zero pattern descriptively rather than as statistical equivalence. For τ, we used an independent-samples t-test to compare exocentric and non-exocentric compounds. Our goal is to evaluate whether semantic similarity provides a quantitative approximation of headedness, so the analysis emphasizes effect direction and magnitude rather than complex model comparison.

3.3. SEMANTIC ASYMMETRY AND HEADEDNESS DIRECTION. The distribution of Δ across headedness patterns is shown in Figure 2. The dashed vertical line marks Δ = 0, where the compound is equally similar to the two constituents. The results broadly support the semantic asymmetry prediction. Left-headed compounds tend to show positive Δ values, right-headed compounds tend to show negative Δ values, and two-headed compounds are distributed closer to zero.

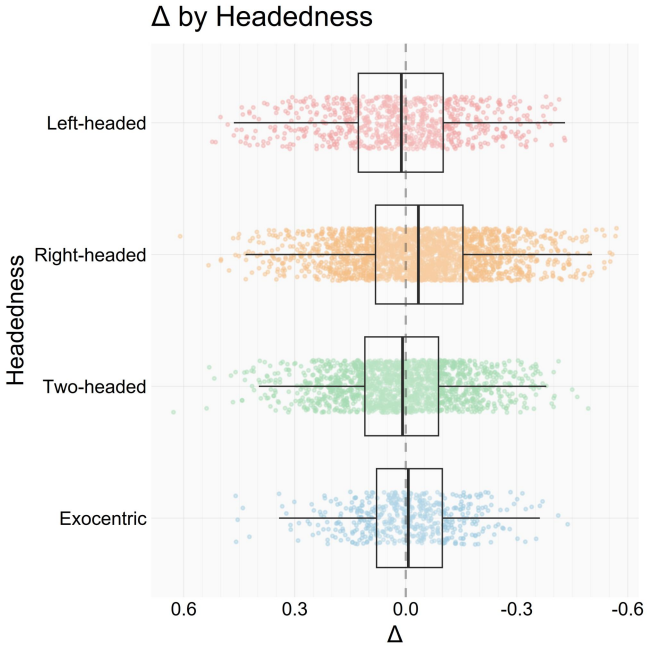


Figure 2. Δ by headedness.

Left-headed compounds show a positive shift in Δ, indicating that the left constituent is, on average, more similar to the whole compound than the right constituent. This pattern is statistically reliable, though small in effect size ($p = .018$, Cohen's $d = 0.09$). Right-headed compounds show the clearest pattern: their Δ values are reliably negative, indicating that the right constituent is more similar to the whole compound ($p < .001$, Cohen's $d = 0.31$). This is the strongest directional effect observed in the analysis.

Two-headed compounds show a weaker pattern. Their Δ values are close to zero, as predicted, but they also show a small positive deviation. This means that Δ reflects two-headedness only weakly. The result is consistent with the idea that coordinate compounds involve more balanced semantic contribution, but it does not show the same clear separation found for right-headed compounds. Exocentric compounds show no reliable directional asymmetry in Δ ($p = .222$, Cohen's $d = 0.07$), which is expected because Δ only measures relative closeness to the left or right constituent. It does not determine whether either constituent is actually close to the compound.

Figure 3 further breaks down the Δ distributions by structural class. The same general pattern is visible, but the strength of the signal differs by class. The right-headed pattern is clearest among ATT compounds, where the modifier–head structure creates a strong semantic asymmetry toward the right constituent. SUB compounds are more heterogeneous, and CRD compounds cluster around zero, as expected for coordinate structures.

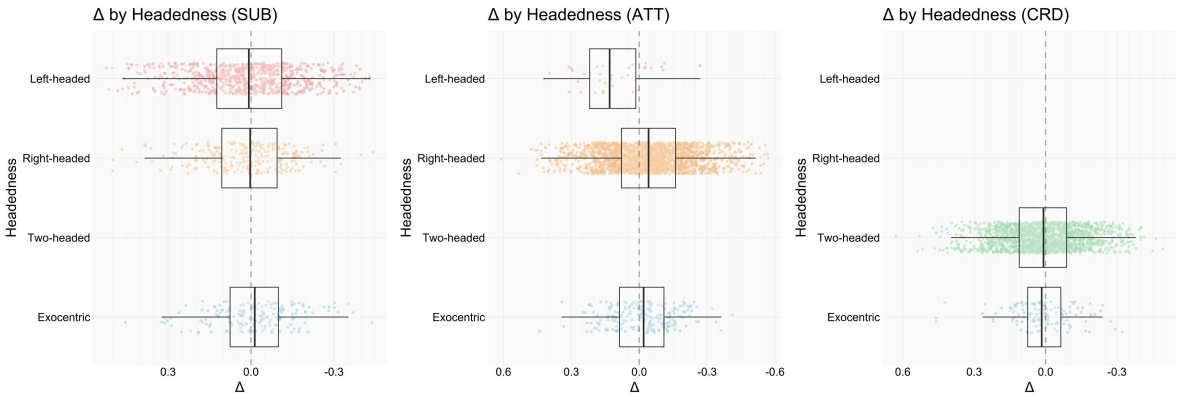


Figure 3. Δ by headedness across structural classes.

These results suggest that Δ captures headedness direction most clearly for right-headed compounds, and to a lesser extent for left-headed compounds. Two-headed compounds are located near the center of the Δ scale, but the effect is weak. Exocentric compounds require a different measure, since their defining property concerns weak anchoring in both constituents rather than left–right asymmetry.

3.4. SEMANTIC PROXIMITY AND EXOCENTRICITY. To test exocentricity, we examined τ, the maximum similarity between the compound and either constituent. If exocentric compounds are semantically less anchored in their internal constituents, they should have lower τ values than non-exocentric compounds.

The results support this prediction. As shown in Figure 4, exocentric compounds have visibly lower τ values than left-headed, right-headed, and two-headed compounds. The difference between exocentric and non-exocentric compounds is statistically reliable and larger than the Δ effects reported above ($p < .001$, Cohen's $d = 0.58$).

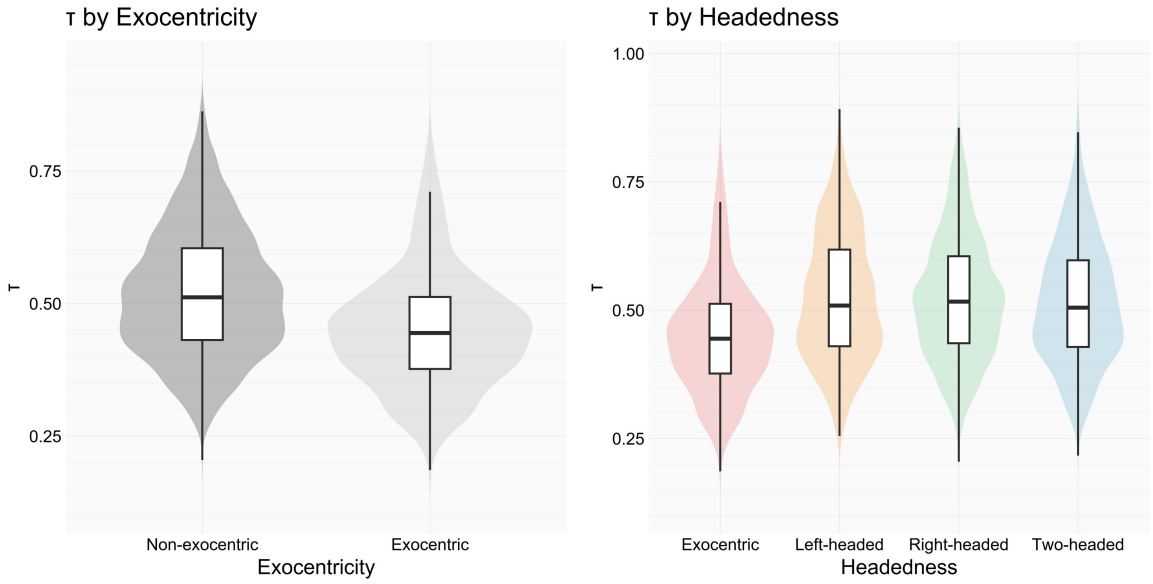


Figure 4. τ by exocentricity and headedness.

This result shows that τ captures a dimension of headedness that Δ does not. Δ measures which constituent is relatively closer to the compound, so it is useful for left–right directionality. τ measures whether the compound is close to at least one constituent, so it is better suited for distinguishing exocentric compounds from endocentric ones. In this sense, exocentricity appears in the embedding space as reduced constituent anchoring: the compound meaning is, on average, more distant from both internal constituents.

Taken together, the two measures provide complementary evidence. Δ captures directional semantic asymmetry, especially for right-headed compounds and more weakly for left-headed compounds. τ captures the lower constituent anchoring of exocentric compounds. Two-headed compounds show the expected tendency toward balanced similarity, though this pattern is weaker than the other effects. These findings suggest that semantic similarity does not replace categorical headedness annotation, but it does provide quantitative support for it: the headedness patterns established in Experiment 1 are reflected as gradient patterns in distributional semantic space.

4. Discussion

Experiment 1 established the empirical baseline for the semantic analyses in Experiment 2. The main class–headedness mapping proposed by Ceccagno and Basciano (2007, 2008) was broadly replicated in a larger high-frequency lexicon: SUB compounds were mostly left-headed, ATT compounds mostly right-headed, and CRD compounds mostly two-headed. This result matters because the semantic similarity analysis depends on the stability of the annotation target. If the headedness patterns were highly specific to a neologism corpus, semantic similarity would have little stable structure to recover. The replication therefore supports the use of the Ceccagno and Basciano system as an empirical framework for testing whether headedness has measurable semantic correlates. At the same time, Experiment 2 shows that these correlates are differentiated. Headedness is reflected in semantic space, but different aspects of headedness are captured by different scores.

4.1. HEADEDNESS AS DIRECTIONAL SEMANTIC ATTRACTION. The semantic asymmetry score (Δ) captures the relative semantic closeness of the compound to its two constituents. In distributional terms, Δ can be interpreted as a measure of directional semantic attraction: a positive score indicates stronger attraction toward the left constituent, a negative score indicates stronger attraction toward the right constituent, and a score near zero indicates balanced similarity. Since distributional semantic models represent lexical meaning through patterns of contextual co-occurrence, this asymmetry should be read as a usage-based trace of how strongly each constituent contributes to the compound's distributional meaning (Baroni & Lenci 2010; Lenci 2008; Turney & Pantel 2010).

The clearest result concerns right-headed compounds. These compounds show a reliable negative Δ, meaning that the compound is more similar to its right constituent than to its left constituent. This pattern is consistent with the structural profile of ATT compounds, which constitute the main source of right-headedness in the dataset. In a typical attributive modifier–head structure, the right constituent provides the core semantic type, while the left constituent restricts or modifies that type. For example, in a compound such as 高考 "college entrance examination", 考 "exam" provides the semantic center, while 高 "high" narrows the interpretation. The strong Δ effect for right-headed compounds suggests that this modifier–head relation is robust enough to leave a measurable trace in embedding space. In this sense, the right-headed pattern is where categorical headedness and distributional similarity align most clearly.

Left-headed compounds also show the predicted direction, but the effect is weaker. This weaker signal is theoretically informative. Left-headed compounds in Chinese are strongly associated with SUB structures, especially verbal and argumental patterns. In such compounds, the left constituent may determine the eventive or relational core of the compound, while the right constituent supplies an internal argument, result, or other semantically salient participant. As a result, the non-head constituent can still contribute substantially to the compound's distributional profile. This helps explain why left-headed compounds show positive Δ values on average, while displaying a less sharply polarized

pattern than right-headed compounds. The weaker effect therefore reflects the internal heterogeneity of SUB compounds rather than a simple failure of the semantic asymmetry score.

The two-headed pattern is more limited. Coordinate compounds are expected to cluster near $\Delta = 0$ because both constituents contribute to the compound's meaning in a relatively balanced way (Bisetto & Scalise 2005; Scalise & Bisetto 2009). This expectation is broadly supported: two-headed compounds are located closer to the center of the Δ scale than right-headed compounds. However, their slight positive deviation from zero shows that coordination does not necessarily produce perfect distributional symmetry. Coordinate compounds may involve synonymy, antonymy, semantic combination, conventional ordering, or frequency asymmetries between constituents. These factors can create small directional biases even when both constituents are morphologically treated as heads. Δ therefore reflects the expected tendency toward balance, but it is a weak diagnostic for two-headedness.

For exocentric compounds, Δ behaves as expected by showing no reliable directional asymmetry. This follows from the nature of the score. Δ only indicates which constituent is relatively closer to the compound. It does not tell us whether either constituent is close to the compound in an absolute sense. An exocentric compound may be slightly closer to the left constituent or slightly closer to the right constituent, while still being semantically distant from both. Therefore, Δ is best understood as a score for headedness direction, especially in endocentric compounds with a unique head. It is not designed to capture the absence of a semantic head.

4.2. EXOCENTRICITY AS REDUCED CONSTITUENT ANCHORING. The semantic proximity score (τ) targets a different dimension of compound structure. While Δ measures left–right asymmetry, τ measures the degree to which the compound remains semantically anchored in at least one constituent. A high τ score means that the compound is close to one of its constituents in semantic space. A low τ score means that the compound is relatively distant from both constituents.

The results show that τ distinguishes exocentric compounds from non-exocentric compounds more strongly than Δ distinguishes headedness direction. Exocentric compounds have lower τ scores than left-headed, right-headed, and two-headed compounds. This pattern fits the traditional definition of exocentricity: an exocentric compound does not denote a subtype of either constituent and lacks an internal semantic head in the relevant morphological sense (Ceccagno & Basciano 2007, 2008; Scalise & Guevara 2006). In distributional terms, this means that the contexts in which the whole compound appears are less similar to the contexts associated with either constituent in isolation.

This result suggests a usage-based interpretation of exocentricity as semantic decoupling. Through lexicalization, metaphorical extension, idiomatization, or pragmatic specialization, the compound may develop a meaning whose distributional profile shifts away from both internal constituents. A compound such as 东西 "east + west =

thing/object" illustrates this point: the whole word does not remain semantically anchored in either "east" or "west". In embedding space, this kind of decoupling should appear as reduced similarity to both constituents, which is exactly what low τ scores capture.

This interpretation connects the present study to work on semantic transparency. Psycholinguistic research treats transparency as a constituent-sensitive property: the meaning of a compound may remain closely related to one or both constituents, or it may become opaque (Libben et al. 2003; Schäfer 2018). Distributional approaches have shown that constituent–compound relations can be modeled through vector-based similarity measures, while also emphasizing that distributional similarity and perceived transparency are related but distinct constructs (Bell & Schäfer 2013, 2016; Wang et al. 2014). In the present study, τ should therefore be interpreted as a distributional proximity score. It measures how strongly the compound shares semantic neighborhoods with its closest constituent. It does not provide a formal definition of exocentricity.

This distinction is important because τ is gradient. Some exocentric compounds may retain a relatively strong connection to one constituent, especially when the compound meaning is metaphorically extended from that constituent. Conversely, some endocentric compounds may show lower τ scores if the compound has become highly conventionalized or if one or both constituents are polysemous. Cosine similarity is also affected by frequency, contextual diversity, and the structure of the embedding space (Günther et al. 2019). For these reasons, τ should be treated as a probabilistic distributional signature of exocentricity rather than a categorical diagnostic.

4.3. COMPLEMENTARY EVIDENCE FROM Δ AND T. The two scores are complementary because they address different questions. Δ asks which constituent the compound is closer to. τ asks whether the compound is close to at least one constituent. A left-headed or right-headed compound should show directional asymmetry in Δ. An exocentric compound should show reduced anchoring in τ. A two-headed compound should show relatively balanced Δ while still maintaining sufficient proximity to its constituents. These are different dimensions of the compound–constituent relationship.

This distinction explains why no single semantic similarity score is sufficient for all headedness patterns. Right-headedness is best captured by Δ because it involves a strong directional bias toward the right constituent. Exocentricity is best captured by τ because it involves weak anchoring in both constituents. Two-headedness is the most difficult pattern to capture because its expected signature is the absence of strong asymmetry. A near-zero Δ is compatible with two-headedness, but it is also compatible with exocentricity, since both may lack a stable left–right directional bias. τ is therefore needed to separate balanced endocentricity from weakly anchored exocentricity.

These findings contribute to the study of Chinese compounding in two ways. First, they support the view that Chinese compounding is strongly shaped by semantic relations among constituents (Song et al. 2022). Second, they refine the long-standing debate over Chinese headedness. Rather than treating Chinese compounds as uniformly right-headed, left-headed, or headless, the results support a more differentiated view: headedness is an

empirical property of individual compounds, constrained by structural class and reflected to varying degrees in semantic space (Ceccagno & Basciano 2007, 2008; Packard 2000). This is also consistent with broader work arguing that distributional semantic models are useful for studying graded meaning, composition, and the interface between lexical semantics and grammar (Boleda 2020; Marelli et al. 2017).

5. Conclusion

This study has shown that headedness distributions in Mandarin Chinese disyllabic compounds are both structurally stable and partially recoverable from compound–constituent semantic similarity. Experiment 1 replicated the main class–headedness mapping proposed by Ceccagno and Basciano (2007, 2008) in a larger high-frequency lexicon. Experiment 2 showed that this annotation-based structure leaves measurable traces in distributional semantic space: the semantic asymmetry score (Δ) captures headedness direction, especially for right-headed compounds, while the semantic proximity score (τ) captures the reduced constituent anchoring of exocentric compounds. Together, these results suggest that semantic similarity provides a meaningful quantitative approximation of headedness when divided into theoretically motivated dimensions. Chinese compound headedness is therefore best understood as a structurally conditioned morphological property that is reflected, to different degrees, in gradient semantic organization.

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